RELATIVE CLASS-NUMBER GROWTH AGAINST RAMIFICATION PRODUCTS FOR CM FIELDS WITH SOLVABLE NORMAL CLOSURES

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Allowed standard prerequisites (editorial).

- CM fields, relative class numbers, ramification-discriminant bounds, Pólya and genus groups. Original cited relative class-number lower bounds and full source-local estimates with complete bibliography retained.

1. Introduction

Let $\mathrm{Cl}(\Delta)$ denote the collection of equivalence classes of primitive binary quadratic forms of discriminant Δ and $h(\Delta) = \#\,\mathrm{Cl}(\Delta)$ be its corresponding class number. In Article 303 of Disquisitiones arithmeticae [11], Gauss conjectured that for every positive integer t, there are only finitely many negative discriminants Δ such that $h(\Delta) = t$. Heilbronn [13] proved this conjecture. In other words, he showed that

$$h(\Delta) \to \infty, \quad \text{as} \quad -\Delta \to \infty.$$

Modifying Heilbronn's method, Chowla [6] proved the following result.

Theorem 1.1 (Chowla). Denote by $\omega(\Delta)$ the number of distinct prime factors of discriminant $\Delta < 0$. Then

$$\frac{h(\Delta)}{2^{\omega(\Delta)}} \to \infty, \quad \text{as} \quad -\Delta \to \infty.$$

Gauss showed that $\mathrm{Cl}(\Delta)$ with an appropriate composition law forms a group. Let the genus $g(\Delta)$ be the cardinality of the 2-torsion part of $\mathrm{Cl}(\Delta)$, i.e., $g(\Delta) = \#\,\mathrm{Cl}(\Delta)[2]$, where

$$\mathrm{Cl}(\Delta)[2] = \{q \in \mathrm{Cl}(\Delta);\ q^2 \sim 1_\Delta\},$$

in which 1_Δ denotes the identity of the group $\mathrm{Cl}(\Delta)$. Gauss proved that $g(\Delta) = 2^{\omega(\Delta)+\epsilon(\Delta)}$, where $\epsilon(\Delta) \in \{0, -1, -2\}$ (see [17, Section 2.2] for the exact formulas for $\epsilon(\Delta)$.) Therefore, Chowla's theorem implies that there are only finitely many negative discriminants Δ for which $h(\Delta) = tg(\Delta)$ for any fixed integer t. This statement was already conjectured, and a complete list of 65 such even discriminants Δ with $g(\Delta) = h(\Delta)$ was proposed by Gauss in the same Article 303. Considering the odd negative discriminants Δ, Dickson [8, pp. 88-89] gave a list of 101 discriminants Δ for which $g(\Delta) = h(\Delta)$.

Equivalently, the above results can be formulated using the terminology of quadratic number fields. For an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-D})$, where $-D$ is a square-free negative integer, let the genus number g_K be the cardinality of the 2-torsion subgroup of the ideal class group $\mathrm{Cl}(K)$ of K. By letting $h_K = \#\,\mathrm{Cl}(K)$, the Dickson's list [8, pp. 88-89] corresponds to 65 square-free negative integers $-D$, given in Table 1, for which $g_K = h_K$. This is classically known as the list of imaginary quadratic fields with "one class in each genus" (for fascinating facts regarding the numbers in this list and their relations with the prime producing idoneal numbers of Euler see the comprehensive survey [17].) Although the completeness of this list still is an open problem, under an assumption on the zeros of the Dirichlet L-function $L(s, \chi_{-D})$, Weinberger [36, Thoeorem 1] proved that this list is complete.

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Theorem 1.2 (Weinberger). There is at most one imaginary quadratic field $K = \mathbb{Q}(\sqrt{-D})$ with $D > 1365$ such that $g_K = h_K$. Moreover, if $L(s, \chi_{-D}) \neq 0$ for $D > 2 \cdot 10^{11}$ and $1 - 1/4 \log D \leqslant s < 1$, then Tabel 1 gives the complete list of imaginary quadratic fields K for which $g_K = h_K$.

$g_K = h_K$	D
1	$1, 2, 3, 7, 11, 19, 43, 67, 163$
2	$5, 6, 10, 13, 15, 22, 35, 37, 51, 58, 91, 115, 123, 187, 235, 267, 403, 427$
4	$21, 30, 33, 42, 57, 70, 78, 85, 93, 102, 130, 133, 177, 190, 195, 253, 435, 483, 555, 595, 627, 715, 795, 1435$
8	$105, 165, 210, 273, 330, 345, 357, 385, 462, 1155, 1995, 3003, 3315$
16	1365

Table 1. The list of 65 imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-D})$ with $D \leqslant 1365$ for which $g_K = h_K$.

The genus theory of quadratic forms can be generalized in various ways to the general setting of number fields. A common generalization in the context of class field theory is given by Fröhlich (see [16] for a detailed treatment.) For a number field K, let $\mathrm{Cl}(K)$ (resp. $\mathrm{Cl}^+(K)$) be the class group (resp. narrow class group) of K, and let h_K (resp. h_K^+) be the class number (resp. narrow class number) of K. Roughly speaking, the genus field (resp. narrow genus field) of K is a specific abelian extension of K whose Galois group is isomorphic to a quotient of $\mathrm{Cl}(K)$ (resp. a quotient of $\mathrm{Cl}^+(K)$). More formally, we have the following definition.

Definition 1.3. For a number field K, the genus field Γ_K (resp. narrow genus field Γ_K^+) of K is the maximal abelian extension of K, which is the compositum of an abelian number field K_* with K and is unramified at all places (resp. at all finite places) of K. The extension degrees $g_K = [\Gamma_K : K]$ and $g_K^+ = [\Gamma_K^+ : K]$ are called the genus number and the narrow genus number of K, respectively.

The analogous problem to the classical one class in each genus for number fields has been studied previously (see [4], [23], and [28]). The family of imaginary quadratic fields is a special case of abelian CM-fields, and a finiteness result, as Theorem 1.1, has been proven for such number fields. Let K be an abelian CM-field, where by an abelian field, we mean a finite Galois extension of $\mathbb{Q}$ whose Galois group is abelian, and by a CM-field we mean a totally complex quadratic extension K of a totally real field K^+ (Indeed, K^+ is the maximal totally real subfield of K.) It can be shown that abelian CM-fields K are the same as imaginary abelian fields (Let K be an imaginary abelian field, i.e., an abelian field with no real embedding. Let σ be the usual complex conjugation. Then σ is an element of order two in the Galois group $G = \mathrm{Gal}(K/\mathbb{Q})$. Let L be the fixed field of the subgroup $\langle\sigma\rangle$. Then $L = K \cap \mathbb{R}$, which implies that L is a totally real Galois field, as G is abelian. Since $[K : L] = 2$, K is an abelian CM-field. The converse is trivial.) As stated before, an example of an abelian CM-field is an imaginary quadratic field. More generally, an imaginary multi-quadratic field, i.e., a field of the form $\mathbb{Q}(\sqrt{-n_1}, \sqrt{-n_2}, \ldots, \sqrt{-n_t})$, for some distinct positive square-free integers $n_1, n_2, \ldots, n_t$, is an abelian CM-field.

The following is proved in [23, Theorem 1].

Theorem 1.4 (Louboutin). There are only finitely many abelian CM-fields K such that $g_K = h_K$.

In fact, Loubotin proves a stronger assertion which gives Theorem 1.4 as a consequence of it. We denote by d_K the absolute value of the discriminant of K. For an abelian CM-field K, let $h_K^- = h_K/h_{K^+}$, where K^+ is the maximal totally real subfield of K. The following generalization of Theorem 1.1 can be extracted from Louboutin's computations in the proof of Theorem 1.4 in [23].

Theorem 1.5. Let K run through an infinite family $\mathcal{F}$ of abelian CM-fields. Then, for every $\epsilon > 0$, we have

$$\frac{(h_K^-)^\epsilon}{\prod_{p|d_K} e_p(K/\mathbb{Q})} \to \infty, \quad \text{as} \quad d_K \to \infty \tag{1.1}$$

in the family $\mathcal{F}$, where $e_p(K/\mathbb{Q})$ denotes the ramification index of a prime p in $K/\mathbb{Q}$.

Note that since $h_K^- \leqslant h_K$ and $g_K \mid \prod_{p|d_K} e_p(K/\mathbb{Q})$ (see Theorem 2.1), then Theorem 1.5 for $\epsilon = 1$ and family $\mathcal{F}$ of abelian CM-fields implies Theorem 1.4.

In this paper, inspired by [10], we are interested in proving results similar to Theorems 1.4 and 1.5 when g_K is replaced with the order of a certain subgroup of $\mathrm{Cl}(K)$. Such a finiteness assertion, similar to Theorem 1.4, is given by Emmelin [10, Corollary 3.15] for the Pólya groups of abelian CM-fields. For a number field K and a prime power q, the ideal

$$\Pi_q(K) =: \prod_{N_{K/\mathbb{Q}}(\mathfrak{p})=q} \mathfrak{p},$$

where $\mathfrak{p}$ denotes a prime ideal of K, is called an Ostrowski ideal. If K has no ideal with norm q, we set $\Pi_q(K) = \mathcal{O}_K$, the ring of integers of K. The following is defined in [2, Chapter II, §3].

Definition 1.6. The Pólya group $\mathrm{Po}(K)$ of a number field K is the subgroup of $\mathrm{Cl}(K)$ generated by the classes of all the Ostrowski ideals $[\Pi_q(K)]$ of K.

From the definition, it is evident that if $K = \mathbb{Q}(\sqrt{D})$ is a quadratic field, then $\mathrm{Po}(K)$ is a 2-group, hence it is a subgroup of $\mathrm{Cl}(K)[2]$. More precisely, for a quadratic field K, we have

$$g_K = \#\,\mathrm{Cl}(K)[2] = \begin{cases} 2^{s_K-2}, & \text{if } D>0,\ N_{K/\mathbb{Q}}(\epsilon_K)=1, \text{ and } D \text{ has a prime divisor } p \equiv 3 \pmod 4, \\ 2^{s_K-1}, & \text{otherwise,} \end{cases} \tag{1.2}$$

and

$$\#\,\mathrm{Po}(K) = \begin{cases} 2^{s_K-2}, & \text{if } D>0, \text{ and } N_{K/\mathbb{Q}}(\epsilon_K)=1, \\ 2^{s_K-1}, & \text{otherwise,} \end{cases} \tag{1.3}$$

where s_K is the number of ramified primes in $K = \mathbb{Q}(\sqrt{D})$, and ϵ_K is the fundamental unit of K when $D > 0$ (see [20, Corollary 9.8 and Theorem 9.10].) Observe that $\#\,\mathrm{Po}(K) = g_K$ for imaginary quadratic fields K, and thus $\#\,\mathrm{Po}(K)$ can be considered as an extension of the classical genus number. Also, $\#\,\mathrm{Po}(K) \mid g_K$ for all quadratic extensions $K/\mathbb{Q}$. However, this is not true in general; for example, for the pure cubic field $K = \mathbb{Q}(\sqrt[3]{20})$, we have $\#\,\mathrm{Po}(K) = h_K = 3$, while $g_K = 1$ (see [22, Example 3.10].)

By Theorem 2.2 below, we see that for Galois extensions $K/\mathbb{Q}$,

$$\#\,\mathrm{Po}(K) \mid \prod_{p|d_K} e_p(K/\mathbb{Q}). \tag{1.4}$$

Thus, [10, Corollary 3.15] can also be obtained from (1.1). In fact, using (1.1) and (1.4), we get the following finiteness theorem for Pólya indices $[\mathrm{Cl}(K) : \mathrm{Po}(K)]$.

Corollary 1.7. For every positive integer t, there are only finitely many abelian CM-fields K with $[\mathrm{Cl}(K) : \mathrm{Po}(K)] = t$.

Remarks 1.8. (i) When $K/\mathbb{Q}$ is Galois, an ideal $\mathfrak{a}$ of $\mathcal{O}_K$ is called ambiguous if $\mathfrak{a}^\sigma = \mathfrak{a}$ for all $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. A class $c \in \mathrm{Cl}(K)$ is called strongly ambiguous if it contains an ambiguous ideal. It can be shown that the Pólya group of a Galois number field K is the same as the group of strongly ambiguous ideal classes of K, see [37, p. 10].

(ii) We can find an example of a family of number fields for which $\mathrm{Po}(K) = \mathrm{Cl}(K)$ for infinitely many K in that family. It is known that for $n \geqslant 3$ and an S_n-field K, i.e., a number field K of degree n whose normal closure has the Galois group isomorphic to the symmetric group S_n, we have $\mathrm{Po}(K) = \mathrm{Cl}(K)$ (see [3, Theorem 4.1] and [25, Theorem 3.10].) On the other hand, by [24, Proposition 6.2], the number of S_n-fields of discriminant less than x is bounded below by $cx^{1/n}$. Hence, in the family of number fields of degree n, there are infinitely many fields K with $\mathrm{Po}(K) = \mathrm{Cl}(K)$.

Our first result establishes versions of (1.1) for families of CM-fields with solvable normal closures.

Theorem 1.9. Let K run through an infinite family $\mathcal{F}$ of CM-fields with solvable normal closures. Write $d_K = d_{K^+}^2 f$, where K^+ denotes the maximal totally real subfield of K.

(i) For every $\epsilon > 0$, we have

$$\frac{(h_K^-)^\epsilon}{\prod_{p|f} e_p(K/\mathbb{Q})} \to \infty, \quad \text{as} \quad d_K \to \infty$$

in the family $\mathcal{F}$.

(ii) If $[K : \mathbb{Q}]$ remains bounded for those fields $K \in \mathcal{F}$ that are not abelian, then

$$\frac{(h_K^-)^\epsilon}{\prod_{p|d_K} e_p(K/\mathbb{Q})} \to \infty, \quad \text{as} \quad d_K \to \infty$$

in the family $\mathcal{F}$.

Moreover, the above results are effective for fields K where $[K : \mathbb{Q}] \neq 2, 4$.

Remark 1.10. The proof of Theorem 1.9 relies on the lower bounds for the class numbers h_K^- given in [32, Theorem 3.1]. Since, under the assumption of Artin's holomorphy conjecture, such bounds hold for any family of CM-fields (see [32, Theorem 3.1]), the assertions of Theorem 1.9 remain true for any family $\mathcal{F}$ of CM-fields under Artin's conjecture.

As a direct consequence of part (ii) of the above theorem and (1.4), we have the following finiteness assertion for solvable CM-fields.

Corollary 1.11. Let K run through an infinite family $\mathcal{F}$ of solvable CM-fields such that $[K : \mathbb{Q}]$ remains bounded for non-abelian fields K in this family. Then, for every positive integer t, there are only finitely many $K \in \mathcal{F}$ with the Pólya index $[\mathrm{Cl}(K) : \mathrm{Po}(K)] = t$.

In light of Corollary 1.11, it is natural to ask for classifying the complete list of abelian CM-fields K or non-abelian solvable CM-fields K of a given degree for which $\mathrm{Po}(K) = \mathrm{Cl}(K)$. In the context of Theorem 1.4 (i.e., "one class in each genus problem"), such a classification for abelian CM-fields is known. In [4], Chang and Kwon classified, unconditionally, all non-quadratic abelian CM-fields K with $g_K = h_K$. In contrast, such classification for abelian CM-fields K with a given Pólya index $[\mathrm{Cl}(K) : \mathrm{Po}(K)]$ seems a challenging task, as computing

$\#\,\mathrm{Po}(K)$ relies on computing the order of the cohomology group $H^1(\mathrm{Gal}(K/\mathbb{Q}), U_K)$ (see Theorem 2.2.) Yet, by studying these cohomology groups for some specific CM-fields, and employing some of the classification results of [4] we establish the following classifications.

Theorem 1.12. In the family of abelian CM-fields, the following assertions hold.

(i) There are 65 imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-D})$, with possibly only one more value D, such that $\mathrm{Po}(K) = \mathrm{Cl}(K)$. The list of such fields coincides with the list given in Table 1.
(ii) There are exactly 57 imaginary bi-quadratic fields $K = \mathbb{Q}(\sqrt{-m}, \sqrt{-n})$ such that $\mathrm{Po}(K) = \mathrm{Cl}(K)$. The complete list of such fields is given in Table 4.
(iii) There are exactly 17 imaginary tri-quadratic fields $K = \mathbb{Q}(\sqrt{-m_1}, \sqrt{-m_2}, \sqrt{-m_3})$ such that $\mathrm{Po}(K) = \mathrm{Cl}(K)$. The complete list of such fields is given in Table 5.
(iv) There are exactly 77 imaginary non-quadratic cyclic fields K such that $\mathrm{Po}(K) = \mathrm{Cl}(K)$. The list of such fields coincides with the list of all imaginary non-quadratic cyclic fields K with $g_K = h_K$ given in [4, Table 1].

In [11, Artcile 303] Gauss conjectures that

$$\Delta = -4d \text{ where } d = 14, 17, 20, 32, 34, 36, 39, 46, 49, 52, 55, 63, 64, 73, 82, 97, 100, 142, 148, 193$$

is the complete list of even negative discriminants Δ with $(h(\Delta), g(\Delta)) = (4, 2)$, i.e., the discriminants of genus two with two classes in each genus. We now know that the above list is complete, as the class number four problem is solved for negative discriminants (see [1]). Inspired by Gauss's list, we conditionally classify imaginary quadratic fields with the Pólya index two.

Theorem 1.13. Under the assumption of the Generalized Riemann Hypothesis (GRH), there are exactly 161 imaginary quadratic fields K with the Pólya index $[\mathrm{Cl}(K) : \mathrm{Po}(K)] = 2$ (equivalently $h_K = 2g_K$). The complete list of such fields is given in Table 6.

As noted before, the finiteness results as Theorem 1.9 (replacing h_K^- with h_K) for families of fields that are not CM may not hold. For example, for any S_n field K we have $\mathrm{Po}(K) = \mathrm{Cl}(K)$. We next, as a direct consequence of the Brauer-Siegel theorem, show that finiteness results similar to Theorem 1.9 (replacing h_K^- with h_K) hold for families of Galois number fields whose degrees and regulators grow modestly with respect to their discriminants.

Theorem 1.14. Let K run through an infinite family $\mathcal{F}$ of Galois number fields. Suppose that

$$\lim_{d_K \to \infty} \frac{[K : \mathbb{Q}]}{\log d_K} = 0, \tag{1.5}$$

and

$$\limsup_{d_K \to \infty} \frac{\log R_K}{\log d_K} < \frac{1}{2} \tag{1.6}$$

in the family $\mathcal{F}$, where R_K denotes the regulator of K. Then, for every $\epsilon > 0$, we have

$$\frac{(h_K)^\epsilon}{\prod_{p \mid d_K} e_p(K/\mathbb{Q})} \to \infty, \quad \text{as} \quad d_K \to \infty \tag{1.7}$$

in the family $\mathcal{F}$.

Remark 1.15. By the Brauer-Siegel theorem, for a family $\mathcal{F}$ of Galois number fields satisfying (1.5), we have

$$\limsup_{d_K \to \infty} \frac{\log R_K}{\log d_K} \leqslant \frac{1}{2}.$$

Thus (1.6) is equivalent to not having $1/2$ as a limit point of the set $\{\log R_K / \log d_K : K \in \mathcal{F}\}$.

It is known that (1.5) holds for any infinite family $\mathcal{F}$ of abelian fields (see [23, Lemma 2].) Thus, we get the following corollary.

Corollary 1.16. The assertion (1.7) holds for a family $\mathcal{F}$ of abelian fields as long as $1/2$ is not a limit point of the set $\{\log R_K/\log d_K : \ K \in \mathcal{F}\}$.

As a concrete example of a family satisfying the conditions of Corollary 1.16, we consider the family of real quadratic fields of extended R-D type.

Definition 1.17. A real quadratic field $K = \mathbb{Q}(\sqrt{D})$ (or simply square-free D) is said to be of Richaud-Degert type, R-D type for short, if $D = \ell^2 + r \neq 5$, where $-\ell < r \leqslant \ell$, and $r \mid 4\ell$. If $r \in \{\pm 1, \pm 4\}$, then D is said to be of narrow R-D type. If we remove the condition $-\ell < r \leqslant \ell$ and include $D = 5$, then D is called extended R-D type.

Since there are infinitely many square-free integers in the form $n^2 + 1$ (see [30]), then the set of extended R-D types is infinite. For real quadratic fields K of extended R-D type, one has $R_K < \log(3d_K)$ (see Lemma 5.5.) Hence, the assertion of Corollary 1.16 holds for this family, and we get the following finiteness result.

Corollary 1.18. For a fixed positive integer t, there are only finitely many real quadratic fields K of extended R-D type with $[\mathrm{Cl}(K) : \mathrm{Po}(K)] = t$.

To illustrate an instance of the above corollary, we classify, with possibly one exception, all real quadratic fields of extended R-D type whose Pólya groups are equal to their ideal class groups.

Theorem 1.19. There are 269 real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type, with possibly only one more value $D \geqslant 6.3 \times 10^{16}$, such that $\mathrm{Po}(K) = \mathrm{Cl}(K)$. These fields are listed in Table 7.

We next observe a close connection between the Pólya indices of real quadratic fields and their genus indices. For a number field K, as before, let h_K^+ and g_K^+ be, respectively, its narrow class number and its narrow genus number. In Corollary 2.5, we show, for a real quadratic field K,

$$\frac{\#\,\mathrm{Po}(K)}{h_K} = \frac{g_K^+}{h_K^+}. \tag{1.8}$$

Hence, for the quadratic fields (imaginary or real), the Pólya index one problem has the same solution as the classical "one class in each genus problem".

In [9], Dohmae classified all real quadratic fields K of narrow R-D type (with possibly only one more field) whose narrow genus numbers g_K^+ are equal to their narrow class numbers h_K^+.

Theorem 1.20 (Dohmae). Let D be of narrow R-D type. Then the following assertions hold.

(i) There are 69 values for D, given in Table 2, such that $D \leqslant 1.31 \times 10^{16}$ and for $K = \mathbb{Q}(\sqrt{D})$ we have $g_K^+ = h_K^+$.
(ii) There is at most one D such that $D > 1.31 \times 10^{16}$ and for $K = \mathbb{Q}(\sqrt{D})$ we have $g_K^+ = h_K^+$.

Our Theorem 1.19 together with (1.8) provides an extension of Theorem 1.20 to real quadratic fields of extended R-D type.

Corollary 1.21. There are 269 real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type, with possibly only one more value $D \geqslant 6.3 \times 10^{16}$, such that $g_K^+ = h_K^+$ (equivalently, $\#\,\mathrm{Po}(K) = h_K$.) The list of such fields is given in Table 7.

$g_K^+ = h_K^+$	D
1	$2, 17, 29, 37, 53, 101, 173, 197, 293, 677$
2	$3, 21, 77, 437$
2	$10, 26, 65, 85, 122, 143, 362, 365, 485, 533, 629,$ $965, 1157, 1685, 1853, 2117, 2813, 3365$
4	$15, 35, 165, 285, 357, 1085, 2397$
4	$170, 290, 530, 957, 962, 1370, 2405, 3485, 9605,$ $10205, 14885, 16133, 20165$
8	$195, 255, 483, 1295, 1365, 2805, 4485, 7917, 8645$
8	$2210, 5330, 32045, 58565, 77285$
16	$1155, 3135, 26565$

Table 2. The list of real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of narrow R-D type with $D \leqslant 1.31 \times 10^{16}$ for which $g_K^+ = h_K^+$.

Finally, by using the results of Theorem 1.19 and solving a Pólya index two problem, we classify, with possibly only one more, all real quadratic fields of extended R-D type whose genus numbers are equal to their class numbers.

Theorem 1.22. There are 298 real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type, with possibly only one more value D, such that $g_K = h_K$. Among them, 269 fields are those K listed in Table 7 with $D < 6.3 \times 10^{16}$ for which $g_K^+ = h_K^+$. The remaining 29 fields K are the ones listed in Table 8 with $D < 4.3 \times 10^{18}$ for which $g_K^+ \neq h_K^+$.

Our proofs employ various lower bounds for the class numbers of number fields. More precisely, the proof of Theorem 1.9 uses the lower bounds for the relative class numbers of abelian CM-fields given by Louboutin in [23, Proof of Theorem 3.1], and the ones provided in Proposition 3.2, derived from Wong [32] and Murty [29]. The lower bound employed in Theorem 1.14 is a direct consequence of the Brauer-Siegel theorem. The proof of Theorem 1.13 uses Ihara's explicit lower bound (conditional on GRH), as given in Theorem 5.3. The classification results for extended R-D types (Theorems 1.19 and 1.22) follow from the explicit lower bound (with one possible exception) given by Mollin and Williams [26, p. 423], derived from Tautuzawa's lower bound [34] for the values of the Dirichlet L-functions at 1. The proof of the classification result of Theorem 1.12 exploits the relations between the order of the Pólya groups and the genus numbers of abelian CM-fields. More precisely, for the fields K in the families considered in Theorem 1.12 it is shown that $\#\,\mathrm{Po}(K) \mid g_K$, and since $g_K \leqslant h_K$, the classification of the Pólya index one problem in these cases reduces to the "one class in each genus problem" for abelian CM-fields which already obtained by Chang and Kwon [4]. Verifying the relation $\#\,\mathrm{Po}(K) \mid g_K$ for abelian CM-fields K in Theorem 1.12 uses a fundamental result of Zantema [37] (see Theorem 2.2) and studying the cohomology groups $H^1(\mathrm{Gal}(K/\mathbb{Q}), U_K)$.

The structure of the paper is as follows. In Section 2 we summarize results on genus fields and Pólya groups and the relation between the orders of the Pólya groups and the genus numbers. The proof of the relation $\#\,\mathrm{Po}(K) \mid g_K$ for imaginary tri-quadratic fields K (Proposition 2.7) is a highlight of this section. The general finiteness theorems (Theorems 1.9 and 1.14) are proved in Section 3. The proof of the unconditional classification of Pólya index one fields for imaginary bi-quadratic and tri-quadratic fields (Theorem 1.12) is given in Section 4. Finally, the details of the conditional classification of imaginary quadratic fields of Pólya index 2 (Theorem 1.13) and the classification results for real quadratic fields of

extended R-D types (Theorems 1.19 and 1.22) are provided in Section 5. We tabulated our classification results in Section 6.

Notation. The following notations will be used throughout this article. For a number field K, the notations $[K:\mathbb{Q}]$, $\mathrm{Cl}(K)$, h_K, h_K^+, g_K, g_K^+, d_K, $\mathcal{O}_K$, U_K, μ_K, and R_K denote the degree, ideal class group, class number, narrow class number, genus number, narrow genus number, absolute value of discriminant, ring of integers, group of units, group of roots of unity, and regulator of K, respectively. By U_K^+ we denote the group of totally positive units of K, and $U_K^2 = \{u^2 : u \in U_K\}$. For a real quadratic field K, we denote the fundamental unit of K by ϵ_K. For a CM-field K with the maximal totally real subfield K^+, the number $h_K^- = h_K/h_{K^+}$ denotes the relative class number of K. Also, we denote the Hasse unit index $[U_K : \mu_K U_{K^+}]$ by W_K. We use $N_{K/\mathbb{Q}}$ to denote both the ideal norm and the element norm map from K to $\mathbb{Q}$. For a Galois extension $K/\mathbb{Q}$, we denote the ramification index of p in $K/\mathbb{Q}$ by $e_p(K/\mathbb{Q})$. The notations $H^i(\mathrm{Gal}(K/\mathbb{Q}), U_K)$ (for an integer $i \geqslant 0$) and $\widehat{H}^0(\mathrm{Gal}(K/\mathbb{Q}), U_K)$ denote the ith cohomology group and the 0th Tate cohomology group of $\mathrm{Gal}(K/\mathbb{Q})$ with the coefficients in U_K, respectively. We denote the cardinality of a finite set S by $\#S$, and the number of divisors of an integer n by $\tau(n)$.

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2. Genus fields and Pólya groups

In this section, we give explicit relations between the genus numbers and the orders of the Pólya groups of some Galois number fields. For genus numbers, we follow the notation and terminology of [16]. Recall that $g_K^+ = [\Gamma_K^+ : K]$ is the narrow genus number of K, where Γ_K^+ is the narrow genus field of K (see Definition 1.3). For abelian number fields, Leopoldt [21] gave the following genus number formula.

Theorem 2.1 (Leopoldt). Let K be an abelian number field. Then

$$g_K^+ = \frac{\prod_{p|d_K} e_p(K/\mathbb{Q})}{[K:\mathbb{Q}]}, \tag{2.1}$$

where $e_p(K/\mathbb{Q})$ denotes the ramification index of a prime p in $K/\mathbb{Q}$.

Recall that the Pólya group $\mathrm{Po}(K)$ of K is the subgroup of $\mathrm{Cl}(K)$ formed by the classes of all Ostrowski ideals of K (see Definition 1.6.) For Pólya groups we use the notation and terminology of [2, Chapter II, §3]. The following fundamental result for Pólya groups of Galois number fields is proved in [37, Section 3].

Theorem 2.2 (Zantema). Let $K/\mathbb{Q}$ be a finite Galois extension with Galois group G. Then the sequence

$$0 \longrightarrow H^1(G, U_K) \longrightarrow \bigoplus_{p|d_K} \frac{\mathbb{Z}}{e_p(K/\mathbb{Q})\mathbb{Z}} \longrightarrow \mathrm{Po}(K) \longrightarrow 0 \tag{2.2}$$

is exact, where $H^1(G, U_K)$ is the first cohomology group of G with coefficients in the unit group U_K of K, and $e_p(K/\mathbb{Q})$ denotes the ramification index of a prime p in $K/\mathbb{Q}$.

As a direct corollary of (2.2), we have

$$\#\,\mathrm{Po}(K) = \frac{\prod_{p|d_K} e_p(K/\mathbb{Q})}{\#H^1(G, U_K)}. \tag{2.3}$$

Combining (2.1) with (2.3), we can explicitly relate the order of the Pólya groups of abelian number fields to their narrow genus numbers.

Proposition 2.3. For an abelian number field K, we have

$$\#\,\mathrm{Po}(K) = \frac{[K:\mathbb{Q}]}{\#H^1(\mathrm{Gal}(K/\mathbb{Q}), U_K)} \cdot g_K^+. \tag{2.4}$$

Proposition 2.3 gives us a key tool to investigate abelian number fields whose Pólya groups are equal to their class groups. In particular, as applications of Formula (2.4), we find that the order of Po(K) divides the narrow genus number of K in some specific families of abelian number fields K.

Proposition 2.4. Let K be a cyclic number field. Then

$$\#\,\mathrm{Po}(K) = \begin{cases} g_K^+/2, & \text{if } K \text{ is real and } N_{K/\mathbb{Q}}(U_K) = \{+1\}, \\ \\ g_K^+, & \text{otherwise.} \end{cases} \tag{2.5}$$

In addition, in the case that K is a real cyclic field, we have

$$\frac{\#\,\mathrm{Po}(K)}{h_K} = \begin{cases} \frac{[U_K^+ : U_K^2]}{2} \cdot \frac{g_K^+}{h_K^+}, & \text{if } N_{K/\mathbb{Q}}(U_K) = \{+1\}, \\ \\ [U_K^+ : U_K^2] \cdot \frac{g_K^+}{h_K^+}, & \text{otherwise,} \end{cases} \tag{2.6}$$

where h_K^+ denotes the narrow class number of K, U_K^+ denotes the totally positive units of K, and

$$U_K^2 = \{u^2 \,:\, u \in U_K\}.$$

Proof. For a cyclic number field K, it is known that $\#H^1(\mathrm{Gal}(K/\mathbb{Q}), U_K) = 2[K:\mathbb{Q}]$ if K is real and $N_{K/\mathbb{Q}}(U_K) = \{+1\}$, and is $[K:\mathbb{Q}]$ otherwise (see [37, Section 3].) Thus, (2.5) follows from (2.4). To obtain (2.6), we use (2.5) and

$$h_K^+ = h_K . \left[U_K^+ : U_K^2\right],$$

see [27, Theorem 1.7]. □

As a consequence of the above proposition, we obtain the following result for real quadratic fields.

Corollary 2.5. If $K = \mathbb{Q}(\sqrt{D})$ is a real quadratic field, for a square-free integer $D > 0$, then

$$\#\,\mathrm{Po}(K) = \begin{cases} g_K/2, & \text{if } N_{K/\mathbb{Q}}(\epsilon_K) = +1 \text{ and } D \text{ has no prime divisor } p \equiv 3 \,(\mathrm{mod}\, 4), \\ g_K, & \text{otherwise,} \end{cases} \tag{2.7}$$

where ϵ_K denotes the fundamental unit of K. Moreover,

$$\frac{\#\,\mathrm{Po}(K)}{h_K} = \frac{g_K^+}{h_K^+}. \tag{2.8}$$

Proof. We have

$$g_K^+ = \begin{cases} 2g_K, & \text{if } D \text{ has a prime divisor } p \equiv 3 \,(\mathrm{mod}\, 4), \\ g_K, & \text{otherwise,} \end{cases} \tag{2.9}$$

see [22, Proposition 3.11]. If D has a prime divisor $p \equiv 3 \,(\mathrm{mod}\, 4)$, then $N_{K/\mathbb{Q}}(\epsilon_K) = +1$. Now by employing (2.9) and Proposition 2.4 we get the relation (2.7) (Note that (2.7) can be also obtained from (1.2) and (1.3).) In addition, using Dirichlet's Unit Theorem, we have

$$4 = \left[U_K : U_K^2\right] = \left[U_K : U_K^+\right] \cdot \left[U_K^+ : U_K^2\right]. \tag{2.10}$$

Now if $N_{K/\mathbb{Q}}(\epsilon_K) = -1$, then $\left[U_K : U_K^+\right] = 4$, since $-1, \epsilon_K \notin U_K \backslash U_K^+$. But, if $N_{K/\mathbb{Q}}(\epsilon_K) = +1$, then $\epsilon_K \in U_K^+$ which implies that $\left[U_K : U_K^+\right] = 2$. Consequently, by (2.10) we obtain

$$\left[U_K^+ : U_K^2\right] = \begin{cases} 2, & \text{if } N_{K/\mathbb{Q}}(\epsilon_K) = +1, \\ 1, & \text{if } N_{K/\mathbb{Q}}(\epsilon_K) = -1. \end{cases} \tag{2.11}$$

From (2.6) and (2.11) we get (2.8). □

We show that the divisibility of g_K by $\#\operatorname{Po}(K)$ also holds for imaginary bi-quadratic and tri-quadratic fields K.

Proposition 2.6. Let K be an imaginary bi-quadratic field. Then

$$\#\operatorname{Po}(K) = \begin{cases} g_K/2, & \text{if } W_K = N_{K^+/\mathbb{Q}}(\epsilon_{K^+}) = 1, \\ \\ g_K, & \text{otherwise,} \end{cases}$$

where K^+ denotes the real quadratic subfield of K, with the fundamental unit ϵ_{K^+}, and $W_K = [U_K : \mu_K \cdot U_{K^+}]$ denotes the Hasse unit index of K in which μ_K is the group of roots of unity in K.

Proof. First, observe that if $K = \mathbb{Q}(\sqrt{-1}, \sqrt{-2})$, then $\#\operatorname{Po}(K) = g_K = h_K = 1$ (Note that in this case, $K^+ = \mathbb{Q}(\sqrt{2})$ has units of negative norm.) So, let $K \neq \mathbb{Q}(\sqrt{-1}, \sqrt{-2})$. Then, from [37, Lemma 4.3], we have

$$H^1(\operatorname{Gal}(K/\mathbb{Q}), U_K) \simeq \begin{cases} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}, & \text{if } W_K = N_{K^+/\mathbb{Q}}(\epsilon_{K^+}) = 1, \\ \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}, & \text{otherwise.} \end{cases}$$

Now the assertion follows from the above isomorphism and Proposition 2.3. □

We next consider the imaginary tri-quadratic fields.

Proposition 2.7. If K is an imaginary tri-quadratic field, then $\#\operatorname{Po}(K) \mid g_K$.

Proof. Let $K = \mathbb{Q}(\sqrt{-m_1}, \sqrt{-m_2}, \sqrt{-m_3})$, where m_1, m_2, m_3 are three distinct square-free positive integers. We show that

$$8 \mid \#H^1(\operatorname{Gal}(K/\mathbb{Q}), U_K).$$

Then the assertion follows from Proposition 2.3.

Let $K^+ = \mathbb{Q}(\sqrt{m_1 m_2}, \sqrt{m_1 m_3})$. Following Setzer's terminology in [35], for $i \in \{0, 1, 2, 3\}$, we say K^+ is of the type M_i if i of the quadratic subfields of K^+ have units of negative norm. For a quadratic subfield k_j of K^+, denote by ϵ_j the fundamental unit of k_j. We have two cases:

(i) At most one of the subfields k_j's has units with negative norm. Hence, K^+ is of the type M_0 or M_1. In this case, by [35, Table I], we find that

$$8 \mid \#H^1(\operatorname{Gal}(K^+/\mathbb{Q}), U_{K^+}). \tag{2.12}$$

Since the cohomology group $H^1(\operatorname{Gal}(K^+/\mathbb{Q}), U_{K^+})$ is embedded in $H^1(\operatorname{Gal}(K/\mathbb{Q}), U_K)$ via the inflation map, then (2.12) yields the claim.

(ii) At least two of the subfields k_j's, say k_1 and k_2, have units with negative norm, i.e., $N_{k_1/\mathbb{Q}}(\epsilon_1) = N_{k_2/\mathbb{Q}}(\epsilon_2) = -1$. Hence, K^+ is of the type M_2 or M_3. From [35, Table I], we observe that either

$$\#H^1(\operatorname{Gal}(K^+/\mathbb{Q}), U_{K^+}) = 8 \quad \text{and} \quad [U_{K^+} : U_{k_1} U_{k_2} U_{k_3}] = 1, \tag{2.13}$$

or

$$\#H^1(\mathrm{Gal}(K^+/\mathbb{Q}), U_{K^+}) = 4 \quad \text{and} \quad [U_{K^+} : U_{k_1}U_{k_2}U_{k_3}] = 2. \tag{2.14}$$

If (2.13) holds, then similar to the case (i), the inflation map implies

$$8 \mid \#H^1(\mathrm{Gal}(K/\mathbb{Q}), U_K).$$

Hence, we assume that (2.14) holds. Since $N_{k_1/\mathbb{Q}}(\epsilon_1) = -1$, the norm map

$$N_{k_1/\mathbb{Q}} : U_{k_1} \to U_{\mathbb{Q}} = \{\pm 1\}$$

is surjective. Consequently, as $k_1/\mathbb{Q}$ is cyclic, by [31, Proposition 1.6.12], we have

$$H^2(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1}) \simeq \widehat{H}^0(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1}) = \frac{U_{\mathbb{Q}}}{N_{k_1/\mathbb{Q}}(U_{k_1})} = \{0\}, \tag{2.15}$$

where $\widehat{H}^0(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1})$ denotes the 0th Tate cohomology group of $\mathrm{Gal}(k_1/\mathbb{Q})$ with coefficient in U_{k_1}. In this case, for the subgroup $\mathrm{Gal}(K/k_1)$ of $\mathrm{Gal}(K/\mathbb{Q})$, we have the following inflation-restriction exact sequence

(2.16)

$$0 \to H^1(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1}) \xrightarrow{\inf} H^1(\mathrm{Gal}(K/\mathbb{Q}), U_K) \xrightarrow{\mathrm{res}} H^1(\mathrm{Gal}(K/k_1), U_K) \to \underbrace{H^2(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1})}_{0}.$$

We first compute the order of $H^1(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1})$. By (2.15), for the cyclic extension $k_1/\mathbb{Q}$, the Herbrand quotient $Q(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1})$ is given by

$$Q(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1}) = \frac{\#\widehat{H}^0(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1})}{\#H^1(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1})} = \frac{1}{\#H^1(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1})}. \tag{2.17}$$

On the other hand, by [5, Proposition 5.10],

$$Q(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1}) = \frac{2^{s_{k_1/\mathbb{Q}}}}{[k_1 : \mathbb{Q}]}, \tag{2.18}$$

where $s_{k_1/\mathbb{Q}}$ denotes the number of infinite places of $\mathbb{Q}$ ramified in k_1. Since k_1 is real, from (2.17) and (2.18), we get

$$\#H^1(\mathrm{Gal}(k_1/\mathbb{Q}), U_{k_1}) = 2. \tag{2.19}$$

Next, we show that

$$4 \mid \#H^1(\mathrm{Gal}(K/k_1), U_K), \tag{2.20}$$

which together with (2.16) and (2.19) imply

$$8 \mid \#H^1(\mathrm{Gal}(K/\mathbb{Q}), U_K),$$

as desired.

We consider two subcases:

(ii-a). Let K^+ be of the type M_2, i.e., $N_{k_3/\mathbb{Q}}(\epsilon_3) = +1$. In this case, to prove (2.20), we show that, in fact,

$$4 \mid \#H^1(\mathrm{Gal}(K^+/k_1), U_{K^+})$$

(Note that $H^1(\mathrm{Gal}(K^+/k_1), U_{K^+})$ is embedded in $H^1(\mathrm{Gal}(K/k_1), U_K)$ via the inflation map.) For the cyclic extension K^+/k_1, the Herbrand quotient $Q(\mathrm{Gal}(K^+/k_1), U_{K^+})$ is given by

$$Q(\mathrm{Gal}(K^+/k_1), U_{K^+}) = \frac{2^{s_{K^+/k_1}}}{[K^+ : k_1]},$$

where s_{K^+/k_1} denotes the number of infinite places of k_1 ramified in K^+. Hence

$$Q(\mathrm{Gal}(K^+/k_1), U_{K^+}) = \frac{1}{2} = \frac{\#\widehat{H}^0(\mathrm{Gal}(K^+/k_1), U_{K^+})}{\#H^1(\mathrm{Gal}(K^+/k_1), U_{K^+})}. \tag{2.21}$$

To deduce that the order of $H^1(\mathrm{Gal}(K^+/k_1), U_{K^+})$ is divisible by 4, we show that

$$\#\widehat{H}^0(\mathrm{Gal}(K^+/k_1), U_{K^+}) > 1.$$

Since

$$\#\widehat{H}^0(\mathrm{Gal}(K^+/k_1), U_{K^+}) = \left[U_{k_1} : N_{K^+/k_1}(U_{K^+})\right],$$

it is enough to show that the norm map

$$N_{K^+/k_1} : U_{K^+} \to U_{k_1}$$

is not surjective. Following [35, p. 161], define

$$\mathcal{C} = \{(u_1, u_2, u_3) \in U_{k_1} \times U_{k_2} \times U_{k_3} : N_{k_1/\mathbb{Q}}(u_1) = N_{k_2/\mathbb{Q}}(u_2) = N_{k_3/\mathbb{Q}}(u_3) = \mathrm{sign}(u_1u_2u_3)\}. \tag{2.22}$$

Then, for the function

$$\begin{aligned} \varphi : U_{K^+} &\to U_{k_1} \times U_{k_2} \times U_{k_3} \\ u &\mapsto \left(N_{K^+/k_1}(u), N_{K^+/k_2}(u), N_{K^+/k_3}(u)\right), \end{aligned}$$

we have

$$\mathrm{Image}(\varphi) \subseteq \mathcal{C}, \tag{2.23}$$

see [35, p. 161]. Now if the norm map N_{K^+/k_1} is surjective, then for the element $u_0 \in U_{K^+}$ with $N_{K^+/k_1}(u_0) = \epsilon_1$ we have

$$\varphi(u_0) = \left(\epsilon_1, N_{K^+/k_2}(u_0), N_{K^+/k_3}(u_0)\right) \in \mathcal{C}.$$

Thus, by (2.22), we get a unit of k_3, say $N_{K^+/k_3}(u_0)$, with negative norm which contradicts the assumption $N_{k_3/\mathbb{Q}}(\epsilon_3) = +1$. Hence, if K^+ is of type M_2, then the norm map

$$N_{K^+/k_1} : U_{K^+} \to U_{k_1}$$

can not be surjective, as claimed. Therefore, by the above argument, (2.20) holds.

(ii-b). Let K^+ be of the type M_3, i.e., $N_{k_3/\mathbb{Q}}(\epsilon_3) = -1$. If the norm map

$$N_{K^+/k_1} : U_{K^+} \to U_{k_1}$$

is not surjective, similar to the case (ii-a), we have $4 \mid \#H^1(\mathrm{Gal}(K^+/k_1), U_{K^+})$ which yields (2.20). So, assume that the norm map N_{K^+/k_1} is surjective. Then,

$$H^2(\mathrm{Gal}(K^+/k_1), U_{K^+}) \simeq \widehat{H}^0(\mathrm{Gal}(K^+/k_1), U_{K^+}) = \{0\}.$$

In this case, from (2.21), we get

$$\#H^1(\mathrm{Gal}(K^+/k_1), U_{K^+}) = 2, \tag{2.24}$$

and for the field extensions $k_1 \subseteq K^+ \subseteq K$, we get the following inflation-restriction exact sequence

$$0 \to H^1(\mathrm{Gal}(K^+/k_1), U_{K^+}) \xrightarrow{\inf} H^1(\mathrm{Gal}(K/k_1), U_K) \xrightarrow{\mathrm{res}} H^1(\mathrm{Gal}(K/K^+), U_K) \to \underbrace{H^2(\mathrm{Gal}(K^+/k_1), U_{K^+})}_{0}. \tag{2.25}$$

Now we show that

$$\#H^1(\mathrm{Gal}(K/K^+), U_K) = 2. \tag{2.26}$$

Then from (2.24) and (2.25), we obtain $\#H^1(\mathrm{Gal}(K/k_1), U_K) = 4$ which implies (2.20).

Since K/K^+ is cyclic, the Herbrand Quotient $Q(\mathrm{Gal}(K/K^+), U_K)$ is given by

$$Q(\mathrm{Gal}(K/K^+), U_K) = \frac{2^{s_{K/K^+}}}{[K:K^+]} = \frac{\#\widehat{H}^0(\mathrm{Gal}(K/K^+), U_K)}{\#H^1(\mathrm{Gal}(K/K^+), U_K)}, \tag{2.27}$$

where s_{K/K^+} denotes the number of infinite places of K^+ ramified in K. Therefore, from (2.27) we get

$$\#H^1(\mathrm{Gal}(K/K^+), U_K) = \frac{1}{8} \cdot \#\widehat{H}^0(\mathrm{Gal}(K/K^+), U_K) = \frac{1}{8} \cdot [U_{K^+} : N_{K/K^+}(U_K)]. \tag{2.28}$$

We next show that

$$[U_{K^+} : N_{K/K^+}(U_K)] = 16,$$

to obtain (2.26) from (2.28). Observe that if $U_K = \mu_K \cdot U_{K^+}$, then by Dirichlet's Unit Theorem, we have

$$[U_{K^+} : N_{K/K^+}(U_K)] = [U_{K^+} : N_{K/K^+}(\mu_K \cdot U_{K^+})] = [U_{K^+} : U_{K^+}^2] = 16,$$

as we desire (Note that since K/K^+ is a quadratic extension and K^+ is totally real, we have $N_{K/K^+}(\mu_K) = \{+1\}$.) Hence, it is enough to show that $W_K = [U_K : \mu_K \cdot U_{K^+}] = 1$. If $\mathbb{Q}(\sqrt{-1}, \sqrt{-2}) \subseteq K$, by [14, Theorem 2], we have $W_K = 1$. Also, if $\mathbb{Q}(\sqrt{-1}, \sqrt{-2}) \nsubseteq K$, then by Lemma 2.8 below, again we have $W_K = 1$ (Recall that we assumed (2.14) holds, in particular the unit index $[U_{K^+} : U_{k_1}U_{k_2}U_{k_3}]$ is 2.) Thus, $[U_{K^+} : N_{K/K^+}(U_K)] = 16$. From here by (2.24), (2.25), and (2.28) we get

$$\#H^1(\mathrm{Gal}(K/k_1), U_K) = 4,$$

which implies that (2.20) holds in this case, too. □

Lemma 2.8. With the above notations, suppose that $N_{k_i/\mathbb{Q}}(\epsilon_{k_i}) = -1$ for every $i = 1, 2, 3$, and $[U_{K^+} : U_{k_1}U_{k_2}U_{k_3}] = 2$. If $\mathbb{Q}(\sqrt{-1}, \sqrt{-2}) \nsubseteq K$, then $W_K = [U_K : \mu_K U_{K^+}]$ equals to 1.

Proof. Since each ϵ_i has a negative norm, by [18, Satz 1], a system of fundamental units for U_{K^+} is given by either $\{\epsilon_{k_1}, \epsilon_{k_2}, \epsilon_{k_3}\}$, or $\{\sqrt{\epsilon_{k_1}\epsilon_{k_2}\epsilon_{k_3}}, \epsilon_{k_2}, \epsilon_{k_3}\}$. Since

$$[U_{K^+} : U_{k_1}U_{k_2}U_{k_3}] = 2,$$

the latter case holds. In particular, $\sqrt{\epsilon_{k_1}\epsilon_{k_2}\epsilon_{k_3}} \in K^+$. By [14, Proof of Theorem 1, Case (4)], if $W_K = 2$ then $K^+(\sqrt{\epsilon_{k_1}\epsilon_{k_2}\epsilon_{k_3}})$ is a quadratic extension of K^+, hence $\sqrt{\epsilon_{k_1}\epsilon_{k_2}\epsilon_{k_3}} \notin K^+$. Therefore, we must have $W_K = 1$. □

3. Proofs of finiteness theorems

The following lower bounds, given by Wong [32], on the relative class numbers h_K^- of CM-fields K with solvable normal closures, play a crucial role in our proofs.

Theorem 3.1. (i) Let K be a CM-field with solvable normal closure and with the maximal totally real subfield K^+. Write $d_K = d_{K^+}^2 f$, and let $n = [K^+ : \mathbb{Q}]$. Then there are effective constants $c_3 > 0$ and $c_4 > 1$ such that

$$h_K^- \geqslant \frac{c_3 c_4^n f^{\left(\frac{1}{2}-\frac{1}{2n}\right)}}{n(2n)^{e(2n)}\delta(2n)}, \tag{3.1}$$

where

$$e(n) = \max_{p^\alpha | n} \alpha, \quad \text{and} \quad \delta(n) = (e(n)+1)^2\, 3^{\frac{1}{3}} 12^{e(n)-1}. \tag{3.2}$$

Moreover, for any $\kappa \in \left[\frac{1}{8\log d_K}, 1\right]$, there exists an effective constant $c(\kappa) > 0$, depending only on κ, such that

$$h_K^- \geqslant \frac{c_3 c(\kappa)^n}{n(2n)^{e(2n)}\delta(2n)} d_{K^+}^{(\frac{1}{2}-\frac{1}{n}-\kappa)} f^{(\frac{1}{2}-\frac{1}{2n})}. \tag{3.3}$$

(ii) The lower bound (3.3) can be improved as

$$h_K^- \geqslant \frac{\tilde{c}_3 c(\kappa)^n}{n(2n)^{e(2n)}\delta(2n)} d_{K^+}^{(\frac{1}{2}-\frac{1}{2n}-\kappa)} f^{(\frac{1}{2}-\frac{1}{4n})}, \tag{3.4}$$

for an ineffective constant $\tilde{c}_3 > 0$.

Proof. For (i) see [32, Theorem 3.1], and for (ii) See [32, Proof of Theorem 1.2]. □

Proposition 3.2. In the family of CM-fields, the following assertions hold.

(i) There exists a constant $C > 0$ such that for any abelian CM-field K with $d_K > C$, we have
$$h_K^- \geqslant d_K^{\frac{1}{16}}.$$
Moreover, if $[K : \mathbb{Q}] \geqslant 6$, then the constant C is effective.

(ii) There exists an ineffective constant $c > 0$ such that for any CM-field $K$with $[K : \mathbb{Q}] = 4$, we have
$$h_K^- \geqslant c d_K^{\frac{1}{12}}.$$

(iii) If K is a CM-field with solvable normal closure, $[K : \mathbb{Q}] = 2n \geqslant 6$, and $d_K = d_{K^+}^2 f$, then there is an effective constant $c(n)$, where $\lim_{n\to\infty} c(n) = \infty$, such that
$$h_K^- \geqslant c(n) f^{\frac{1}{3}}.$$

(iv) If K is a CM-field with solvable normal closure such that $[K : \mathbb{Q}] = 2n \geqslant 6$, then there is an effective constant $b(n) > 0$, such that
$$h_K^- \geqslant b(n) d_K^{\frac{1}{24}}.$$

Proof. (i) See [23, Proof of Theorem 1].

(ii) If $[K : \mathbb{Q}] = 4$, then $[\tilde{K} : \mathbb{Q}] \leqslant 24$, where $\tilde{K}$ denotes the normal closure of K over $\mathbb{Q}$. Hence $\tilde{K}/\mathbb{Q}$ is a solvable extension. Write $d_K = d_{K^+}^2 f$. Then, using the lower bound (3.4) for $n = 2$ and $\kappa = \frac{1}{12}$, there is an ineffective constant $c > 0$ such that, for $d_K \geqslant e^{\frac{3}{2}}$,

$$h_K^- \geqslant c d_{K^+}^{\frac{1}{6}} f^{\frac{3}{8}} \geqslant c d_K^{\frac{1}{12}}.$$

(iii) By the lower bound (3.1) given in Theorem 3.1, we have

$$h_K^- \geqslant c(n) f^{\left(\frac{1}{2}-\frac{1}{6}\right)} = c(n) f^{\frac{1}{3}},$$

where

$$c(n) := \frac{c_3 c_4^n}{n(2n)^{e(2n)}\delta(2n)}.$$

Note that $c_4 > 1$ and by (3.2), $e(n) \leqslant \log n / \log 2$. Hence, $\lim_{n\to\infty} c(n) = \infty$.

(iv) Let $\kappa \in \left[\frac{1}{8\log d_K}, 1\right]$ be given. Employing $d_K = d_{K^+}^2 f$ in (3.3) we get

$$h_K^- \geqslant \frac{c_3 c(\kappa)^n}{n(2n)^{e(2n)}\delta(2n)} d_K^{(\frac{1}{4}-\frac{1}{2n}-\frac{\kappa}{2})} f^{(\frac{1}{4}+\frac{\kappa}{2})}. \tag{3.5}$$

For $\kappa = \frac{1}{12}$, from (3.5), we get

$$h_K^- \geqslant b(n) d_K^{\frac{1}{24}}, \quad \text{for } d_K \geqslant e^{3/2},$$

where

$$b(n) = \frac{c_3 c(1/12)^n}{n(2n)^{e(2n)}\delta(2n)}. \tag{3.6}$$

□

We are now ready to prove our first result.

Proof of Theorem 1.9. (i) Let $M, \epsilon > 0$ and for $K \in \mathcal{F}$ set

$$\frac{(h_K^-)^\epsilon}{\prod_{p|f} e_p(K/\mathbb{Q})} < M. \tag{3.7}$$

We consider cases. If $[K : \mathbb{Q}] = 4$, then by employing part (ii) of Proposition 3.2 in (3.7), and $d_K = d_{K^+}^2 f$, we get

$$\prod_{p|d_{K^+}} p^{\frac{m_p\epsilon}{16}} \prod_{p|f} \frac{p^{\frac{m_p\epsilon}{16}}}{e_p(K/\mathbb{Q})} \leqslant \frac{(h_K^-)^\epsilon}{\prod_{p|f} e_p(K/\mathbb{Q})} < M, \tag{3.8}$$

where $d_K = \prod_{p|d_K} p^{m_p}$. By [19, Chapter III, Proposition 8] we have $e_p(K/\mathbb{Q}) \leqslant 2m_p$. Hence, from (3.8) we have

$$f(d_K) := \prod_{p|d_{K^+}} p^{\frac{m_p\epsilon}{16}} \prod_{p|f} \frac{p^{\frac{m_p\epsilon}{16}}}{2m_p} \leqslant \frac{(h_K^-)^\epsilon}{\prod_{p|f} e_p(K/\mathbb{Q})} < M. \tag{3.9}$$

By [12, Theorem 316], $f(d_K) \to \infty$, as $d_K \to \infty$. Hence, from (3.9) we conclude that the number of $K \in \mathcal{F}$ with $[K : \mathbb{Q}] = 4$ satisfying (3.7) is finite.

Using a similar argument and employing part (i) of Proposition 3.2, we have that the number of $K \in \mathcal{F}$ with $[K : \mathbb{Q}] = 2$ satisfying (3.7) is finite (Note that in this case, $d_{K^+} = 1$ as $K^+ = \mathbb{Q}$.)

Now assume that $2n = [K : \mathbb{Q}] \geqslant 6$, $K \in \mathcal{F}$, and (3.7) holds. Then, from part (iii) of Proposition 3.2, we have

$$c(n)^\epsilon \prod_{p|f} \frac{p^{\frac{m_p\epsilon}{3}}}{e_p(K/\mathbb{Q})} \leqslant \frac{(h_K^-)^\epsilon}{\prod_{p|f} e_p(K/\mathbb{Q})} < M.$$

Since $c(n) \to \infty$, as $n \to \infty$, and $e_p(K/\mathbb{Q}) \leqslant 2m_p$ implies the product over f is bounded below by some positive constant, we conclude that the degrees n for such K's are bounded. Now from part (iv) of Proposition 3.2 and (3.7), we get

$$b(n)^\epsilon \prod_{p|d_{K^+}} p^{\frac{m_p\epsilon}{24}} \prod_{p|f} \frac{p^{\frac{m_p\epsilon}{24}}}{e_p(K/\mathbb{Q})} \leqslant \frac{(h_K^-)^\epsilon}{\prod_{p|f} e_p(K/\mathbb{Q})} < M,$$

where $b(n)$ is the function given by (3.6). Applying $e_p(K/\mathbb{Q}) \leqslant 2m_p$ in the above inequality yields

$$j(d_K) := \prod_{p|d_{K^+}} p^{\frac{m_p\epsilon}{24}} \prod_{p|f} \frac{p^{\frac{m_p\epsilon}{24}}}{2m_p} \leqslant \frac{(h_K^-)^\epsilon}{b(n)^\epsilon \prod_{p|f} e_p(K/\mathbb{Q})} < \frac{M}{b(n)^\epsilon}. \tag{3.10}$$

As above, since by [12, Theorem 316], $j(d_K) \to \infty$, as $d_K \to \infty$, and $b(n)$ is bounded, as n is bounded, then (3.10) shows that, the number of $K \in \mathcal{F}$ with $[K : \mathbb{Q}] \geqslant 6$ satisfying (3.7) is finite.

Hence, for given $M, \epsilon > 0$, there are only finitely many $K \in \mathcal{F}$ satisfying (3.7). Thus, the claimed assertion holds.

(ii) Similar to part (i), for $M, \epsilon > 0$ and $K \in \mathcal{F}$ set

$$\frac{(h_K^-)^\epsilon}{\prod_{p|d_K} e_p(K/\mathbb{Q})} < M \tag{3.11}$$

and consider cases.

Suppose that there are infinitely many CM-fields K in the family $\mathcal{F}$ such that $K/\mathbb{Q}$ is not abelian and $2n = [K : \mathbb{Q}]$ remains bounded as $d_K \to \infty$. Note that for such fields $K \in \mathcal{F}$, we must have $[K : \mathbb{Q}] = 2n \geqslant 4$. Now, for $2n \geqslant 6$, by employing part (iv) of Proposition 3.2 and $e_p(K/\mathbb{Q}) \leqslant 2m_p$, from (3.11) we get

$$r(d_K) := \prod_{p|d_K} \frac{p^{\frac{m_p\epsilon}{24}}}{2m_p} \leqslant \frac{(h_K^-)^\epsilon}{b(n)^\epsilon \prod_{p|d_K} e_p(K/\mathbb{Q})} < \frac{M}{b(n)^\epsilon}. \tag{3.12}$$

Since, by [12, Theorem 316], $r(d_K) \to \infty$ as $d_K \to \infty$, and $b(n)$ is bounded, as n is bounded, then (3.12) shows that, the number of non-abelian CM-fields $K \in \mathcal{F}$ satisfying (3.11) is finite.

An identical argument which uses part (i) (respectively part (ii)) of Proposition 3.2, instead of part (iv), establishes the finiteness of abelian (respectively quartic non-Galois) fields in the family $\mathcal{F}$ satisfying (3.11). □

Proof of Theorem 1.14. Since (1.5) holds, by Brauer-Siegel's theorem,

$$\lim_{d_K\to\infty} \frac{\log(h_K R_K)}{\log d_K} = \frac{1}{2}. \tag{3.13}$$

Let $\eta > 0$ be such that

$$\limsup_{d_K\to\infty} \frac{\log R_K}{\log d_K} < \frac{1}{2} - 2\eta. \tag{3.14}$$

(Note that by (1.6) such η exists.) Hence, from (3.13) and (3.14), there exists a $C_\eta > 0$ such that for $d_K > C_\eta$ we have

$$\frac{1}{2} - \eta < \frac{\log(h_K R_K)}{\log d_K} < \frac{\log h_K}{\log d_K} + \frac{1}{2} - 2\eta.$$

Thus, for $d_K > C_\eta$, we have $h_K > {d_K}^\eta$. Now, let $d_K = \prod_{p|d_K} p^{m_p}$. By [19, Chapter III, Proposition 8] we have $e_p(K/\mathbb{Q}) \leqslant 2m_p$. Then, for a given $\epsilon > 0$ and K in the family $\mathcal{F}$ with $d_K > C_\eta$, we have

$$\frac{(h_K)^\epsilon}{\prod_{p|d_K} e_p(K/\mathbb{Q})} > \prod_{p|d_K} \frac{(p^{m_p})^{\eta\epsilon}}{e_p(K/\mathbb{Q})} \geqslant \prod_{p|d_K} \frac{(p^{m_p})^{\eta\epsilon}}{2m_p}.$$

By [12, Theorem 316] the right-hand side of the above inequality approaches ∞ as $d_K \to \infty$. Thus, the result holds. □

4. Proof of Theorem 1.12

Proof. (i) By part (i) of Proposition 2.4, we have $\#\operatorname{Po}(K) = g_K$ for an imaginary quadratic field K. Now the assertion follows from Theorem 1.2.

(ii) By Proposition 2.6, we have

$$\#\operatorname{Po}(K) = \begin{cases} g_K/2, & \text{if } W_K = N_{K^+/\mathbb{Q}}(\epsilon_{K^+}) = 1, \\ \\ g_K, & \text{otherwise,} \end{cases} \tag{4.1}$$

which implies $\#\operatorname{Po}(K) \leqslant g_K$. Since $g_K \leqslant h_K$, the number fields in [4, Table III] (also in [28, Table 2]), consisting of all imaginary bi-quadratic fields K with $g_K = h_K$, form an

m	n	$\#\,\mathrm{Po}(K)$	$g_k = h_K$
37,58	163	1 by [33, Theorem 5]	2
67	123,235,403	1 by [33, Theorem 5]	2
91	163	1 by [33, Theorem 5]	2
91	403	1 by [33, Theorem 7]	2
115	163	1 by [33, Theorem 5]	2
115	235	1 by [33, Theorem 7]	2
163	187,235,267,403	1 by [33, Theorem 5]	2
1	30, 42, 70, 78, 190	1 by [33, Lemma 6]	4
13	78	1 by [33, Theorem 5]	4

Table 3. The list of imaginary bi-quadratic fields $K = \mathbb{Q}(\sqrt{-m}, \sqrt{-n})$ with $g_K = h_K$ for which $\#\,\mathrm{Po}(K)$ is computed by using the results of [33].

initial list of all such K with $\mathrm{Po}(K) = \mathrm{Cl}(K)$. Now it suffices to exclude those fields K from [4, Table III] for which $\#\,\mathrm{Po}(K) = g_K/2$, and for the remaining cases we will have $\#\,\mathrm{Po}(K) = h_K$. By (4.1), the desired list is formed by all the fields K in [4, Table III] for which $W_K \neq N_{K^+/\mathbb{Q}}(\epsilon_{K^+})$. To check this inequality, we use PARI/GP[1] to compute W_K and $N_{K^+/\mathbb{Q}}(\epsilon_{K^+})$ and compare them. For the cases given in Table 3, the software was not able to compute $N_{K^+/\mathbb{Q}}(\epsilon_{K^+})$ for which we use some results of [33], as stated in the third column of Table 3, to compute $\#\,\mathrm{Po}(K)$. For all these cases, we conclude that $\#\,\mathrm{Po}(K) \neq h_K$.

(iii) The list given in [4, Table V] (or [28, Table 3]) provides the complete classification of all imaginary tri-quadratic fields K with $g_K = h_K$. We observe that $h_K = 1$ for any field K in this list. On the other hand, by Proposition 2.7, the order of $\mathrm{Po}(K)$ divides the genus number g_K, hence $\#\,\mathrm{Po}(K) \leqslant g_K \leqslant h_K$. Consequently, Table 5 gives the list of all imaginary tri-quadratic fields with $\mathrm{Po}(K) = \mathrm{Cl}(K)$.

(iv) The complete classification of all imaginary non-quadratic cyclic fields K with $g_K = h_K$ is given in [4, Table 1]. Part (i) of Proposition 2.4 shows that this is also the complete classification of all imaginary non-quadratic cyclic fields whose Pólya groups are equal to their class groups. □

5. Proofs of other classification theorems

We start by expressing the order of the Pólya group of a quadratic field in terms of the divisor function.

Lemma 5.1. Let $K = \mathbb{Q}(\sqrt{D})$ be a quadratic field, where D is a square-free integer. Let

$$c_K = \begin{cases} 1/2, & \text{if } D \equiv 1 \pmod 4, \\ 1/3, & \text{if } D \equiv 3 \pmod 4, \\ 1/4, & \text{if } D \equiv 2 \pmod 4, \end{cases}$$

and denote by τ the number of divisors function. Then,

$$\#\,\mathrm{Po}(K) = \begin{cases} \frac{1}{2} c_K \tau(d_K), & \text{if } D > 0 \text{ and } K \text{ has no units of negative norm}, \\ c_K \tau(d_K), & \text{otherwise.} \end{cases} \tag{5.1}$$

[1] https://pari.math.u-bordeaux.fr/

Proof. For $n=\prod_{i=1}^{k} p_i^{\alpha_i}$, we have $\tau(n)=\prod_{i=1}^{k}(\alpha_i+1)$. Thus,

$$\tau(d_K)=\begin{cases} 2^{s_K}, & \text{if } D\equiv 1 \pmod 4,\\ 3\cdot 2^{s_K-1}, & \text{if } D\equiv 3 \pmod 4,\\ 2^{s_K+1}, & \text{if } D\equiv 2 \pmod 4,\end{cases}$$

where s_K denotes the number of ramified primes in $K/\mathbb{Q}$. Now the assertion follows from (1.3). □

In the next lemma we establish an explicit upper bound for the divisor function.

Lemma 5.2. Let $n=2^{\beta}\prod_{i=1}^{r} q_i$ be a positive integer, where q_i's are distinct odd prime numbers and β is a non-negative integer. Then

$$\frac{\tau(n)}{\sqrt[4]{n}} \leqslant \begin{cases} 2.8908, & \text{if } \beta=0,\\ \left(\frac{\beta+1}{\sqrt[4]{2}}\right)\cdot 2.8908, & \text{if } \beta\geqslant 1.\end{cases} \tag{5.2}$$

In particular, for a quadratic field $K=\mathbb{Q}(\sqrt{D})$, we have

$$\tau(d_K) < c'_K \sqrt[4]{d_K}, \tag{5.3}$$

where

$$c'_K=\begin{cases} 2.8908, & \text{if } D\equiv 1 \pmod 4,\\ 7.2927, & \text{if } D\equiv 3 \pmod 4,\\ 9.7235, & \text{if } D\equiv 2 \pmod 4.\end{cases}$$

Proof. Since for each prime $p\geqslant 17$, we have $\frac{2}{\sqrt[4]{p}}<1$, then

$$\frac{\tau(n)}{\sqrt[4]{n}} \leqslant \begin{cases} \displaystyle\prod_{\substack{p \text{ prime},\\ 2<p<17}} \frac{2}{\sqrt[4]{p}}, & \text{if } \beta=0,\\ \frac{\beta+1}{\sqrt[4]{2^{\beta}}} \displaystyle\prod_{\substack{p \text{ prime},\\ 2<p<17}} \frac{2}{\sqrt[4]{p}}, & \text{if } \beta\geqslant 1.\end{cases}$$

The relation (5.2) follows by noting that 2.8908 is an upper bound for the product in the above inequality. The bound (5.3) is an immediate consequence of (5.2). □

5.1. Proof of Theorem 1.13. To prove Theorem 1.13, we use Lemmas 5.1 and 5.2 along with the following conditional explicit lower bound, proved in [15, Theorem 5], for the class numbers of imaginary quadratic fields.

Theorem 5.3 (Ihara). Let K be an imaginary quadratic field with the absolute value discriminant d_K. Under GRH, if $\alpha_K:=\frac{1}{2}\log d_K > 1.16$ (i.e., if $d_K\geqslant 11$), then

$$h_K > \frac{\frac{\pi}{6}\sqrt{d_K}-\alpha_K+b_1}{\alpha_K+2\log\alpha_K+b_2+c(\alpha_K)}, \tag{5.4}$$

with b_1, b_2, $c(\alpha_K)$ are given by

$$\begin{cases} b_1=2\log M+\log 2-4q_0=-0.34037-4q_0,\\ b_2=2\log M-2\gamma_{\mathbb{Q}}+\log 2+1=-0.49480\ldots,\\ c(t)=\frac{4\log t+2}{t-1},\end{cases}$$

where $q_0 = e^{-\pi\sqrt{d_K}}$, $M = 0.596450134\ldots$, $\log M = -0.516759638\ldots$, and $\gamma_{\mathbb{Q}}$ is the Euler-Kronecker constant.

Now, we are ready to prove Theorem 1.13.

Proof of Theorem 1.13. By Lemmas 5.1 and 5.2 and Theorem 5.3, we have

$$\frac{2\cdot \#\mathrm{Po}(K)}{h_K} < \frac{c_K c_K' \sqrt[4]{d_K}\left(\frac{1}{2}\log d_K + 2\log\log\sqrt{d_K} + b_2 + \frac{4\log\log\sqrt{d_K}+2}{\log\sqrt{d_K}-1}\right)}{\frac{\pi}{6}\sqrt{d_K} - \frac{1}{2}\log d_K + b_1}, \tag{5.5}$$

where b_1, b_2 are the constants given in Theorem 5.3.

We define, for $K = \mathbb{Q}(\sqrt{-D})$ and $x > 0$, the function $f_K(x)$ as

$$f_K(x) = \frac{c_K c_K' \sqrt[4]{x}\left(\log\sqrt{x} + 2\log\log\sqrt{x} + b_2 + \frac{4\log\log\sqrt{x}+2}{\log\sqrt{x}-1}\right)}{\frac{\pi}{6}\sqrt{x} - \log\sqrt{x} + b_1}.$$

We observe that

$$\begin{cases} f_K(x) < 1, & \text{if } x \geqslant 3.6\times 10^7 \text{ and } -D \equiv 1 \pmod 4, \\ \\ f_K(x) < 1, & \text{if } x \geqslant 4.1\times 10^8 \text{ and } -D \equiv 2,3 \pmod 4. \end{cases}$$

Hence, by (5.5), if $[\mathrm{Cl}(K) : \mathrm{Po}(K)] = 2$, under GRH, we must have

$$\begin{cases} d_K < 3.6\times 10^7, & \text{if } -D \equiv 1 \pmod 4, \\ \\ d_K < 4.1\times 10^8, & \text{if } -D \equiv 2,3 \pmod 4. \end{cases} \tag{5.6}$$

Using PARI/GP[2] we compute h_K and $\#\mathrm{Po}(K)$ for all imaginary quadratic fields K with d_K smaller than the bound (5.6) to obtain the classification given in Table 6. □

5.2. Proof of Theorem 1.19. We need the following three lemmas.

Lemma 5.4. If $K = \mathbb{Q}(\sqrt{\ell^2 + r})$ is a real quadratic field of R-D type, then

$$u_K := \begin{cases} \ell + \sqrt{\ell^2 + r}, & \text{if } |r| = 1, \\ \\ \frac{\ell+\sqrt{\ell^2+r}}{2}, & \text{if } |r| = 4, \\ \\ \frac{2\ell^2+r+2\ell\sqrt{\ell^2+r}}{|r|}, & \text{if } |r| \neq 1, 4, \end{cases} \tag{5.7}$$

is the fundamental unit of K. More generally, u_K is a unit if K is of extended R-D type.

Proof. If K is of R-D type, by [7, Theorem 1], u_K is the fundamental unit of K. Since u_K has norm ± 1, we only need to show that u_K is an algebraic integer when K is of extended R-D type. Let D be an extended R-D type integer which is not of R-D type. Then $D = \ell^2 + r$, where $r \mid 4\ell$ and either $r > \ell$ or $r \leqslant -\ell$. If $|r| = 4$, then for $r = 4$ (resp. $r = -4$), $\ell \in \{1, 2, 3\}$ (resp. $\ell \in \{3, 4\}$.) It is straightforward to check that u_K is an algebraic integer in these cases. If $|r| \neq 4$, then either $(\ell, r) = (1, 2)$ or $\ell > 1$ and $r \in \{-\ell, \pm 2\ell_1, \pm 4\ell_1\}$ for some divisor ℓ_1 of ℓ. For $(\ell, r) = (1, 2)$, we have $D = 3$. In this case, $u_K = 2 + \sqrt{3}$ is an algebraic integer. Now let $\ell > 1$. If $r = \pm 4\ell_1$, then $D = \ell^2 \pm 4\ell_1 \equiv 1 \pmod 4$ (Note that in this case ℓ cannot be even, as D is square-free.) Thus,

$$\frac{2\ell^2 + r + 2\ell\sqrt{\ell^2 + r}}{|r|} = \frac{(\ell^2/\ell_1 \pm 2) + (\ell^2/\ell_1)\sqrt{D}}{2}$$

[2]https://pari.math.u-bordeaux.fr/

is an algebraic integer in K. Finally, if $r = -\ell, \pm 2\ell_1$, then r divides both $2\ell^2 + r$ and 2ℓ which implies that

$$\frac{2\ell^2 + r + 2\ell\sqrt{\ell^2 + r}}{|r|} \in \mathbb{Z}[\sqrt{D}].$$

□

The following assertion is stated in [26, p. 423].

Lemma 5.5. Let $K = \mathbb{Q}(\sqrt{D})$ be a real quadratic field of extended R-D type. Then $R_K < \log(3D)$, where R_K denotes the regulator of K.

Proof. As usual, denote by ϵ_K the fundamental unit of K. First observe that for $D = 5$, we have $\epsilon_K = \frac{-1+\sqrt{5}}{2}$. So, $R_K = \log(\epsilon_K) \approx 0.482 < \log(3D) \approx 2.7$.

Next, let $D = \ell^2 + r \neq 5$ with $r \mid 4\ell$. We have three cases:
Case 1. If $|r| = 1$, then by Lemma 5.4,

$$\epsilon_K = \ell + \sqrt{\ell^2 \pm 1} < \big(2\ell^2 \pm 2\big) + \big(\ell^2 \pm 1\big) = 3D.$$

(Note that $\ell \geqslant 2$ for $r = -1$.)
Case 2. If $|r| = 4$, then, by Lemma 5.4,

$$u_K = \frac{\ell + \sqrt{\ell^2 + r}}{2}$$

is a unit of K. Hence,

$$\epsilon_K \leqslant \frac{\ell + \sqrt{\ell^2 \pm 4}}{2} < \big(2\ell^2 \pm 8\big) + \big(\ell^2 \pm 4\big) = 3D.$$

(Note that $\ell \geqslant 3$ for $r = -4$.)
Case 3. if $|r| \neq 1, 4$, then by Lemma 5.4,

$$u_K = \frac{2\ell^2 + r + 2\ell\sqrt{\ell^2 + r}}{|r|}$$

is a unit of K. If $\ell = 1$, the only possibility is to have $r = 2$, i.e., $D = 3$. In this case, $\epsilon_K = 2 + \sqrt{3}$ and $R_K = \log \epsilon_K \approx 1.31 < \log(3D) \approx 2.2$.

Now assume that $\ell > 1$. Since $r \leqslant 4\ell \leqslant 2\ell^2$, then $\ell\sqrt{\ell^2 + r} < 2\ell^2$.
For $r > 0$,

$$\epsilon_K \leqslant \frac{2\ell^2 + r + 2\ell\sqrt{\ell^2 + r}}{|r|} \leqslant \frac{2\ell^2 + r + 2\ell\sqrt{\ell^2 + r}}{2} < \left(\ell^2 + \frac{r}{2}\right) + 2\ell^2 < 3\ell^2 + 3r = 3D.$$

For $r < 0$, we have

$$\epsilon_K \leqslant \frac{2\ell^2 + r + 2\ell\sqrt{\ell^2 + r}}{|r|} < \frac{2\ell^2 + 2\ell\sqrt{\ell^2}}{2} = 2\ell^2. \tag{5.8}$$

Since $r \mid 4\ell$, for $\ell \geqslant 12$, we get $-3r \leqslant 12\ell \leqslant \ell^2$. Consequently, from (5.8) we find that for $\ell \geqslant 12$, the inequality $\epsilon_K < 2\ell^2 \leqslant 3\ell^2 + 3r = 3D$ holds. We numerically check that the relation

$$\frac{2\ell^2 + r + 2\ell\sqrt{\ell^2 + r}}{|r|} < 3\ell^2 + 3r = 3D$$

also holds for $1 \leqslant \ell \leqslant 11$ and $r \mid 4\ell$ with $r \neq \pm 1, \pm 4$. □

The following explicit lower bound for the class numbers of quadratic fields of extended R-D type is a consequence of the analytic class number formula, Lemma 5.5, and Tatuzawa's lower bound [34] for the values of Dirichlet L-functions at 1.

Lemma 5.6. Let $K = \mathbb{Q}(\sqrt{D})$ run through the family of real quadratic fields of extended R-D type. Then, with only one possible exception, we have

$$h_K > \frac{0.655\, d_K^{7/16}}{32 \log(3D)}. \tag{5.9}$$

Proof. See [26, p. 423] for a proof. □

We are now ready to prove Theorem 1.19.

Proof of Theorem 1.19. Replacing (5.4) with (5.9), we follow the same method as in the proof of Theorem 1.13. For $K = \mathbb{Q}(\sqrt{D})$, we consider two cases:
Case 1. If $N_{K/\mathbb{Q}}(\epsilon_K) = -1$, then by Lemmas 5.1, 5.2, and 5.6, we have

$$\frac{\#\,\mathrm{Po}(K)}{h_K} < \frac{32 c_K c_K' \log(3D)}{0.655\, d_K^{3/16}} \leqslant \frac{32 c_K c_K' \log(3 d_K)}{0.655\, d_K^{3/16}}. \tag{5.10}$$

Observe that for the function

$$f_K(x) := \frac{32 c_K c_K' \log 3x}{0.655\, x^{3/16}}, \quad x > 0,$$

we have

$$\begin{cases} f_K(x) < 1, & \text{if } x \geqslant 4.3 \times 10^{18} \text{ and } D \equiv 1 \pmod 4, \\ \\ f_K(x) < 1, & \text{if } x \geqslant 8.14 \times 10^{19} \text{ and } D \equiv 2, 3 \pmod 4. \end{cases}$$

Consequently, for a real quadratic field $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type, with one possible exception, if K has some units of negative norm and $\mathrm{Po}(K) = \mathrm{Cl}(K)$, then

$$\begin{cases} d_K < 4.3 \times 10^{18}, & \text{if } D \equiv 1 \pmod 4, \\ \\ d_K < 8.14 \times 10^{19}, & \text{if } D \equiv 2, 3 \pmod 4. \end{cases} \tag{5.11}$$

Case 2. Let K has no units of negative norm, i.e., $N_{K/\mathbb{Q}}(\epsilon_K) = +1$. In this case, by an argument similar to Case 1, we find that if $\mathrm{Po}(K) = \mathrm{Cl}(K)$, then

$$\begin{cases} d_K < 6.3 \times 10^{16}, & \text{if } D \equiv 1 \pmod 4, \\ \\ d_K < 1.3 \times 10^{18}, & \text{if } D \equiv 2, 3 \pmod 4. \end{cases} \tag{5.12}$$

Using PARI/GP[3], we compute h_K and $\#\,\mathrm{Po}(K)$ for all real quadratic fields K of extended R-D type for which the discriminant d_K satisfies in (5.11) or (5.12) to get Table 7. □

5.3. Proof of Theorem 1.22.

Proof of Theorem 1.22. Let $K = \mathbb{Q}(\sqrt{D})$ be a real quadratic field of extended R-D type such that $g_K = h_K$. We consider two cases:
Case 1. If $\#\,\mathrm{Po}(K) = h_K$, i.e., if K is any of those fields given in Table 7, then $\#\,\mathrm{Po}(K) = g_K = h_K$. This is true, since $g_K \leqslant h_K$ and by Corollary 2.5, $\#\,\mathrm{Po}(K) \mid g_K$.
Case 2. If $\#\,\mathrm{Po}(K) \neq h_K$, then from $g_K = h_K$ and Corollary 2.5, we get

$$\#\,\mathrm{Po}(K) = g_K/2 = h_K/2. \tag{5.13}$$

[3]https://pari.math.u-bordeaux.fr/

Moreover, K has no units of negative norm and D has no prime divisor $p \equiv 3 \pmod 4$. Now, by an argument similar to the proof of Theorem 1.19, we deduce that in this case, if (5.13) holds, then

$$\begin{cases} d_K < 4.3 \times 10^{18}, & \text{if } D \equiv 1 \pmod 4, \\ \\ d_K < 8.14 \times 10^{19}, & \text{if } D \equiv 2, 3 \pmod 4. \end{cases} \tag{5.14}$$

We use PARI/GP[3] to list all real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type with $h_K = 2^{s_K - 1}$ such that K has no units of negative norm, D has no prime divisor $p \equiv 3 \pmod 4$, and the discriminant d_K is bounded by (5.14) (Note that by (1.3), $\#\operatorname{Po}(K) = 2^{s_K - 2}$, and by (5.13), we get $g_K = h_K = 2^{s_K - 1}$.) Using these computations and Corollary 2.5, we get the list, given in Table 8, of all real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type, with $D < 4.3 \times 10^{18}$, such that $g_K = h_K$ and $g_K^+ \neq h_K^+$. □

6. Tables

$\#\,\mathrm{Po}(K) = h_K$	m	n
1	1	$2, 3, 5, 7, 11, 13, 19, 37, 43, 67, 163$
	2	$3, 7, 10, 11, 19, 37, 43, 58, 67$
	3	$6, 7, 11, 15, 19, 43, 51, 67, 123, 163, 267$
	7	$11, 19, 35, 43, 91, 163, 247$
	11	$19, 22, 67.163, 187$
	19	$67, 163$
	43	$67, 163$
	67	163
2	1	$6, 10, 58$
	2	$5, 6, 13, 22, 37$
	5	10

Table 4. The list of all imaginary bi-quadratic fields $K = \mathbb{Q}(\sqrt{-m}, \sqrt{-n})$ with $\mathrm{Po}(K) = \mathrm{Cl}(K)$.

$\#\,\mathrm{Po}(K) = h_K$	(m_1, m_2, m_3)
1	$(1,2,5), (1,2,3), (1,3,5), (1,2,11), (3,5,7),$ $(3,5,2), (1,5,7), (3,2,11), (7,5,2), (1,13,7),$ $(3,17,11), (1,3,7), (1,3,11), (3,7,2),$ $(1,3,19), (1,7,19), (3,11,19)$

Table 5. The list of all imaginary tri-quadratic fields $K = \mathbb{Q}(\sqrt{-m_1}, \sqrt{-m_2}, \sqrt{-m_3})$ with $\mathrm{Po}(K) = \mathrm{Cl}(K)$.

$h_K = 2 \cdot \#\,\mathrm{Po}(K)$	D
4	$14, 17, 34, 39, 46, 55, 73, 82, 97, 142, 155, 193, 203, 219, 259,$ $291, 323, 355, 667, 723, 763, 955, 1003, 1027, 1227, 1243,$ $1387, 1411, 1507, 1555$
8	$65, 66, 69, 77, 114, 138, 141, 145, 154, 205, 213, 217, 238, 258,$ $265, 282, 301, 310, 322, 418, 438, 442, 445, 498, 505, 553, 598,$ $651, 658, 697, 742, 793, 915, 987, 1131, 1443, 1635, 1659,$ $1771, 1947, 2035, 2067, 2139, 2163, 2451, 2667, 2715, 2755,$ $3243, 3355, 3507, 4123, 4323, 5083, 5467, 6307$
16	$285, 390, 429, 465, 510, 561, 570, 609, 645, 690, 777, 798, 805,$ $858, 870, 897, 910, 957, 1005, 1045, 1065, 1105, 1110, 1113,$ $1122, 1185, 1290, 1302, 1353, 1605, 1645, 1653, 1677, 1705,$ $1870, 1885, 2002, 2013, 2170, 2233, 2737, 3795, 4515, 5115,$ $5187, 6195, 7035, 7315, 7395, 7755, 7995, 8547, 8715, 8835,$ $9867, 11067, 11715, 13195, 14763, 16555$
32	$1785, 2145, 2310, 2730, 3045, 3570, 3705, 4305, 4830, 4845,$ $5005, 19635, 31395, 33915, 40755$

Table 6. The list of all the imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-D})$ with $[\mathrm{Cl}(K) : \mathrm{Po}(K)] = 2$ (Under GRH).

$g_K = \#\operatorname{Po}(K) = h_K$	$g_K^+ = h_K^+$	D
1	1	$2, 5, 13, 17, 29, 37, 53, 101, 173, 197, 293, 677$
1	2	$3, 6, 7, 11, 14, 21, 23, 33, 38, 47, 62, 69, 77, 83, 93, 141,$ $167, 213, 227, 237, 398, 413, 437, 453, 573, 717, 1077,$ $1133, 1253, 1293, 1757$
2	2	$10, 26, 65, 85, 122, 362, 365, 485, 533, 629, 965, 1157,$ $1685, 1853, 2117, 2813, 3365$
2	4	$15, 30, 35, 39, 42, 51, 66, 78, 87, 95, 102, 105, 110, 119,$ $123, 138, 143, 165, 182, 203, 215, 222, 230, 258, 285, 287,$ $318, 327, 357, 395, 402, 429, 447, 527, 623, 635, 645, 678,$ $741, 782, 843, 885, 902, 957, 1022, 1085, 1173, 1245, 1298,$ $1533, 1605, 1965, 2013, 2037, 2085, 2093, 2301, 2373,$ $2397, 2613, 2717, 3237, 3597, 3605, 3813, 4245, 4277,$ $4773, 4893, 5757, 5885, 5957, 6573, 7733, 14405$
4	4	$170, 290, 530, 962, 1370, 2405, 3485, 9605, 10205, 14885,$ $16133, 20165$
4	8	$195, 210, 231, 255, 330, 390, 435, 455, 462, 483, 570, 615,$ $627, 663, 770, 798, 903, 915, 930, 1095, 1190, 1218, 1230,$ $1235, 1295, 1302, 1365, 1463, 1482, 1515, 1518, 1547, 1595,$ $1610, 1722, 1767, 1770, 1938, 2015, 2030, 2387, 2595, 2607,$ $2618, 2805, 2910, 3045, 3230, 3335, 3723, 3885, 4389, 4485,$ $4758, 5565, 6045, 6405, 6765, 7293, 7917, 8645, 9933,$ $10005, 10965, 11165, 12045, 13485, 13845, 14685, 15645,$ $16653, 17765, 19565, 20405, 21045, 24045, 30597, 31317,$ $33117, 41613$
8	8	$2210, 5330, 32045, 58565, 77285$
8	16	$11155, 1995, 2310, 2415, 2730, 3003, 3135, 3255, 3570,$ $3927, 3990, 4290, 4935, 5187, 5610, 5655, 6090, 6555, 7035,$ $7215, 7755, 9030, 10010, 12155, 12558, 13695, 14630,$ $21318, 23205, 26565, 35805, 74613, 108885$
16	32	$19635, 25935, 33495, 451605$

Table 7. The list of all real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type, with $D < 6.3 \times 10^{16}$, for which $g_K = \#\operatorname{Po}(K) = h_K$. This is also the list of all such fields K with $g_K^+ = h_K^+$.

$g_K = 2 \cdot \#\operatorname{Po}(K) = h_K$	D
2	$34, 146, 194, 205, 221, 482, 1205, 1469, 1517, 2045, 3005, 5645,$ $7157, 8333, 9005$
4	$410, 890, 3842, 3965, 7565, 7685, 18245, 25493, 41093, 55205$
8	$10370, 19610, 22490$
16	81770

Table 8. The list of all real quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of extended R-D type, with $D < 4.3 \times 10^{18}$, such that $g_K = h_K$ and $g_K^+ \neq h_K^+$.

References

[1] Steven Arno, The imaginary quadratic fields of class number 4, Acta Arith. 60 (1992), no. 4, 321–334, DOI 10.4064/aa-60-4-321-334. MR1159349

[2] Paul-Jean Cahen and Jean-Luc Chabert, Integer-valued polynomials, Mathematical Surveys and Monographs, vol. 48, American Mathematical Society, Providence, RI, 1997. MR1421321

[3] Jean-Luc Chabert and Emmanuel Halberstadt, From Pólya fields to Pólya groups (II): Non-Galois number fields, J. Number Theory 220 (2021), 295–319, DOI 10.1016/j.jnt.2020.06.008. MR4177545

[4] Ku-Young Chang and Soun-Hi Kwon, The imaginary abelian number fields with class numbers equal to their genus class numbers, J. Théor. Nombres Bordeaux 12 (2000), no. 2, 349–365, DOI 10.5802/jtnb.283 (English, with English and French summaries). Colloque International de Théorie des Nombres (Talence, 1999). MR1823189

[5] Nancy Childress, Class field theory, Universitext, Springer, New York, 2009. MR2462595

[6] S. Chowla, An extension of Heilbronn's class-number theorem, The Quarterly Journal of Mathematics 1 (1934), 304–307.

[7] Günter Degert, Über die Bestimmung der Grundeinheit gewisser reell-quadratischer Zahlkörper, Abh. Math. Sem. Univ. Hamburg 22 (1958), 92–97, DOI 10.1007/BF02941943 (German). MR0092824

[8] Leonard Eugene Dickson, Introduction to the Theory of Numbers, University of Chicago Press, 1929.

[9] Kazuhiro Dohmae, On real quadratic fields with a single class in each genus, Japan. J. Math. (N.S.) 19 (1993), no. 2, 241–250, DOI 10.4099/math1924.19.241. MR1265653

[10] Étienne Emmelin, A note on Pólya groups, arXiv:2302.07977, https://arxiv.org/abs/2302.07977 (2023), 11 pages.

[11] Carl Friedrich Gauss, Disquisitiones arithmeticae, Yale University Press, New Haven, Conn.-London, 1966. Translated into English by Arthur A. Clarke, S. J. MR0197380

[12] G. H. Hardy and E. M. Wright, An introduction to the theory of numbers, 6th ed., Oxford University Press, Oxford, 2008. Revised by D. R. Heath-Brown and J. H. Silverman; With a foreword by Andrew Wiles. MR2445243

[13] Hans Heilbronn, On the class-number in imaginary quadratic fields, The Quarterly Journal of Mathematics 1 (1934), 150–160.

[14] Mikihito Hirabayashi and Ken-ichi Yoshino, Unit indices of imaginary abelian number fields of type $(2, 2, 2)$, J. Number Theory 34 (1990), no. 3, 346–361, DOI 10.1016/0022-314X(90)90141-D. MR1049510

[15] Yasutaka Ihara, On the Euler-Kronecker constants of global fields and primes with small norms, Algebraic geometry and number theory, Progr. Math., vol. 253, Birkhäuser Boston, Boston, MA, 2006, pp. 407–451, DOI 10.1007/978-0-8176-4532-8_5. MR2263195

[16] Makoto Ishida, The genus fields of algebraic number fields, Lecture Notes in Mathematics, vol. Vol. 555, Springer-Verlag, Berlin-New York, 1976. MR0435028

[17] Ernst Kani, Idoneal numbers and some generalizations, Ann. Sci. Math. Québec 35 (2011), no. 2, 197–227 (English, with English and French summaries). MR2917832

[18] Tomio Kubota, Über den bizyklischen biquadratischen Zahlkörper, Nagoya Math. J. 10 (1956), 65–85 (German). MR0083009

[19] Serge Lang, Algebraic number theory, Addison-Wesley Publishing Co., Inc., Reading, Mass.-London-Don Mills, Ont., 1970. MR0282947

[20] Franz Lemmermeyer, Quadratic number fields, Springer Undergraduate Mathematics Series, Springer, Cham, [2021] ©2021. Translated from the 2017 German original. MR4331428

[21] Heinrich W. Leopoldt, Zur Geschlechtertheorie in abelschen Zahlkörpern, Math. Nachr. 9 (1953), 351–362, DOI 10.1002/mana.19530090604 (German). MR0056032

[22] Amandine Leriche, About the embedding of a number field in a Pólya field, J. Number Theory 145 (2014), 210–229, DOI 10.1016/j.jnt.2014.05.002. MR3253301

[23] Stéphane Louboutin, A finiteness theorem for imaginary abelian number fields, Manuscripta Math. 91 (1996), no. 3, 343–352, DOI 10.1007/BF02567959. MR1416716

[24] Gunter Malle, On the distribution of Galois groups, J. Number Theory 92 (2002), no. 2, 315–329, DOI 10.1006/jnth.2001.2713. MR1884706

[25] Abbas Maarefparvar, On Pólya groups of non-Galois number fields, To appear in Functiones et Approximatio, Commentarii Mathematici. arXiv: 2408.09019 https://www.arxiv.org/abs/2408.09019 (2025), 20 pages.

[26] R. A. Mollin and H. C. Williams, Solution of the class number one problem for real quadratic fields of extended Richaud-Degert type (with one possible exception), Number theory (Banff, AB, 1988), de Gruyter, Berlin, 1990, pp. 417–425. MR1106676
[27] J. S. Milne, Class Field Theory (v4.03), 2020. Available at www.jmilne.org/math/.
[28] Ichiro Miyada, On imaginary abelian number fields of type $(2, 2, \cdots, 2)$ with one class in each genus, Manuscripta Math. 88 (1995), no. 4, 535–540, DOI 10.1007/BF02567840. MR1362937
[29] V. Kumar Murty, Class numbers of CM-fields with solvable normal closure, Compositio Math. 127 (2001), no. 3, 273–287, DOI 10.1023/A:1017589432526. MR1845038
[30] Trygve Nagel, Zur Arithmetik der Polynome, Abh. Math. Sem. Univ. Hamburg 1 (1922), no. 1, 178–193, DOI 10.1007/BF02940590 (German). MR3069398
[31] Jürgen Neukirch, Alexander Schmidt, and Kay Wingberg, Cohomology of number fields, 2nd ed., Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], vol. 323, Springer-Verlag, Berlin, 2008. MR2392026
[32] Peng-Jie Wong, On Stark's class number conjecture and the generalised Brauer-Siegel conjecture, Bull. Aust. Math. Soc. 106 (2022), no. 2, 288–300, DOI 10.1017/S0004972721001076. MR4476085
[33] Mohammed Taous and Abdelkader Zekhnini, Pólya groups of the imaginary bicyclic bi-quadratic number fields, J. Number Theory 177 (2017), 307–327, DOI 10.1016/j.jnt.2017.01.025. MR3629246
[34] Tikao Tatuzawa, On a theorem of Siegel, Japanese journal of mathematics: transactions and abstracts, 1952, pp. 163–178.
[35] C. Bennett Setzer, Units over totally real $C_2 \times C_2$ fields, J. Number Theory 12 (1980), no. 2, 160–175, DOI 10.1016/0022-314X(80)90049-9. MR0578808
[36] P. J. Weinberger, Exponents of the class groups of complex quadratic fields, Acta Arith. 22 (1973), 117–124, DOI 10.4064/aa-22-2-117-124. MR0313221
[37] H. Zantema, Integer valued polynomials over a number field, Manuscripta Math. 40 (1982), no. 2-3, 155–203, DOI 10.1007/BF01174875. MR0683038